

on silicon carbide (GaN-on-SiC) process technology, which offers superior efficiency, power density and affordability.

The device enables engineers to design higher power radar solutions with significant size, weight, power, and cost (SWAP-C) improvements and bring them to market faster.

Qorvo

www.qorvo.com

DC-DC controllers for vehicles

Renesas Electronics Corporation has announced a pair of new 42V dual synchronous controllers with low 6µA quiescent current and integrated 2A source/3A sink MOSFET drivers that provide primary power stage solutions for Renesas R-Car H3 and R-Car M3 SoCs.

The ISL78264 dual sync buck controller manages DC-DC conversion by stepping down the 12V battery system to 5V and 3.3V to power the vehicle's electronic control unit (ECU), in-vehicle infotainment (IVI) and digital cockpit systems with 50W to 200W power levels.

The ISL78263 dual sync boost and buck controller provides DC-DC conversion to support power levels of 25W to 100W while providing a pre-boost when the battery voltage (VBAT) dips to as low as 2.1V during a cranking transient or a start/stop event.

Renesas

www.renesas.com

Railway DC-DC converters

TRACO POWER introduces TEP 40UIR and TEP 60UIR, two series of high-performance railway DC-DC converters with ultra-wide 12:1 input voltage range.

The ultra-wide input allows the converter to act as an all-in-one solution for covering different voltage ranges in the same application, thereby resolving the issue of having multiple different converters installed.

An internal circuit present in these modules helps to extend the hold-up time with ease while eliminating the

need for expensive high voltage capacitors to cover the full input range. With only a 25V capacitor (independent of the input voltage), the whole input range can be covered, reducing cost, size, and inrush current.

All models are approved for railway applications according to EN 50155 and EN 61373. These are also qualified according to EN 45545-2 (for fire behaviour).

TRACO POWER

www.tracopower.com

TEST & MEASUREMENT

Oscilloscope

Tektronix has announced the expansion of its TBS1000 series with the launch of the new TBS1000C digital storage oscilloscope. It has a 17.7cm (7-inch) WVGA colour display with up to 1 GS/s sample rate and bandwidths from 50MHz to 200MHz. It also includes Tektronix's HelpEverywhere system, which provides useful tips and hints.

The TBS1000C digital storage oscilloscope has been designed for quick hands-on learning and easy operation.



Besides a WVGA colour display, a network of ten vertical divisions and fifteen horizontal divisions enables users to see more signals per screen.

The series has dedicated front panel controls to provide easy access to important settings and new large menus with clearly labelled on-screen information for easily finding the required data.

Tektronix

in.tek.com

Vector signal generator

Rohde & Schwarz has released a new multi-standard and fully configurable platform, R&S SMCV100B vector signal generator for automotive, broadcast, navigation, and wireless applications, ranging from the lab to production. The R&S SMCV100B can cover the 5G NR extended FR1 frequency range of up to 7.125GHz,

making it ideal for mobile communications applications.



In the automotive industry, the R&S SMCV100B is suited for end-of-line testing of car radios or GNSS navigation equipment. Broadcast standards can be combined with mobile communications, wireless, and navigation standards. For Go/NoGo tests, the generator can offer GPS, GLONASS, Galileo, and BeiDou signals of a single navigation satellite.

Rohde & Schwarz

www.rhodeschwarz.com

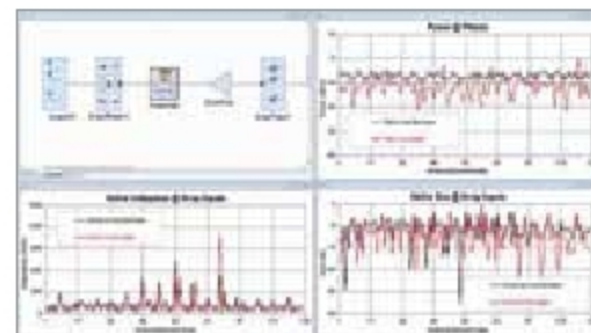
SOFTWARE

PathWave Design software

Keysight Technologies has announced the release of PathWave Design 2021. The latest version is an open, scalable, and predictable 5G and mmWave software solution.

Through this release, design and validation engineers will be able to accelerate the delivery of chip, board, and system products by integrating device, circuit, and system design with improved performance and accuracy.

Keysight's PathWave Design software suite includes PathWave Advanced Design System (ADS), PathWave RFIC Design (GoldenGate), PathWave System



Design (SystemVue), and PathWave EM Design (EMPro), which deliver the key capabilities needed to address 5G challenges such as radio frequency/microwave (RF/MW) simulation and verification, electronic system-level (ESL) simulation and device modelling.

Keysight Technologies

www.keysight.com